Thesis Econometrics and Data Science

Evaluating the RVU levy exemption using AI-generated survey responses

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Abstract

This thesis evaluates the RVU levy exemption, a Dutch regulation designed to help employees with demanding jobs retire up to 3 years early. As data on demanding work is not readily available, we use machine learning to predict whether exemption users had demanding work in the pre-exemption era. To this end, we select five questions from the LISS Work & Schooling survey related to different aspects of demanding work and use Machine Learning to predict responses based on CBS microdata. Using these predictions, we assess the fraction of exemption users who had demanding work in 2020 and whether they are over- or underrepresented compared to 1) the employed population and 2) the exemption-eligible population. Since assessments can only be made when misclassification rates are known, and true values are unavailable for exemption users and control groups, assumptions must be made about the transportability of estimated misclassification rates to predictions. A benchmark analysis, assuming nondifferentiality, suggests that the exemption targets individuals who had irregular working times and work involving certain aspects of physical load. However, a robustness analysis and a Bayesian importance sampling procedure show that these conclusions depend heavily on the informativeness of estimates regarding true misclassification rates. Hence, further research is required to confidently evaluate the RVU levy exemption.

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LISS panel

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CBS

Results based on calculations by SEO Amsterdam Economics in project number 3081 using non-public microdata from Statistics Netherlands. Under certain conditions, these microdata are accessible for statistical and scientific research. For further information: Microdata: Conducting your own research | CBS.

UWV

I would like to thank the Uitvoeringsinstituut Werknemersverzekeringen (UWV) for providing the dataset on RVU users and for allowing me to use it for my thesis, which was essential to this research.

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1 Introduction

1.1 Motivation and research question

To increase labour market participation among older citizens, the statutory retirement age (SRA) in the Netherlands, also referred to as the AOW age, has increased stepwise from 65 to 67 years in recent years. In the coming years, the SRA was set to increase by 8 months for every additional year in life expectancy. However, in January 2026, a new coalition agreed to increase the SRA more quickly than originally planned, to 12 months per additional year in life expectancy. In addition, early retirement policies have become stricter, giving fewer people the opportunity to retire before the SRA (Visser et al., 2016). Where increasing the SRA reduces public costs and increases public income, concerns have been raised about the impact on people with demanding jobs, as many already struggle to reach retirement in good health (Vermeer et al., 2016) and some enter social support programmes (Chirikos and Nestel, 1991).

To partially alleviate this issue, a new early retirement policy was introduced in 2021: the RVU levy exemption. This regulation offers fiscal benefits when retiring up to 3 years before the SRA and is intended to support employees who, due to the heavy nature of their work, are unable to reach the SRA in good health (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a). Since legislators deemed it impossible to determine at a central level what constitutes heavy work (Tweede Kamer der Staten-Generaal, 2019), the exemption was implemented as a general tax measure, and it was left to employers and employees to determine who can use an RVU, either through a collective or an individual agreement.

Originally, the exemption was scheduled to run until the end of 2025, but following the agreement *Gezond naar het Pensioen*, it will be continued from 2026 onward, with the aim of becoming structural (Ministerie van Sociale Zaken en Werkgelegenheid, 2025c). The exemption will remain a general tax measure, but continuation is contingent on RVUs being used more targeted and controlled during CAO negotiations (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a). Starting in 2026, TNO will ex ante evaluate whether RVU possibilities in CAOs align with the intended goal of supporting employees with demanding work, and will annually evaluate how RVUs are used in CAO agreements.

Policy makers are often interested in the effectiveness and efficiency of a new policy, where effectiveness refers to how well the intended target is reached, and efficiency refers to how efficiently the allocated resources are used to achieve it. An important determinant of the effectiveness and efficiency is whether the policy is used by the intended target audience or also by people for whom it is not meant (Ernst et al., 2025). For the RVU levy exemption, the targeted audience consists of employees who cannot reach the SRA in good health due to the heavy nature of their work.

Although employers bear the primary financial burden of an RVU, misuse of the exemption can both reduce labour supply in an already tight labour market and lead to missed taxes (Tweede

Kamer der Staten-Generaal, 2019). Therefore, it is important to evaluate whether RVUs are used by the targeted groups. However, research into RVU usage since the introduction of the exemption is limited.

Ministerie van Sociale Zaken en Werkgelegenheid (2025b) investigates characteristics of RVU users in the period 2021-2024 and shows that in 2024, male users and slightly higher middle incomes are overrepresented. Ministerie van Sociale Zaken en Werkgelegenheid (2025d) investigates requirements for RVUs as agreed in CAOs in January 2025, and shows that out of 273 CAOs with a collective RVU, 36 offered the possibility of an RVU to everyone, and 63 only required a few years of service. Moreover, almost no CAOs include objective criteria for classifying work as demanding. While these studies provide some insights into whether the exemption worked as intended, the former does not relate to heavy work, and the latter does not examine actual RVU use but only the possibilities offered through collective agreements.

Hence, to our knowledge, no research has assessed the RVU levy exemption by investigating the relationship between RVU use and demanding work. Therefore, this thesis aims to answer the question: How well did the RVU levy exemption target employees with demanding jobs in the period 2021-2024?

To answer this question, we will assess what fraction of RVU users in the period 2021-2024 had jobs related to several aspects of demanding work before the exemption was introduced, and whether they are over- or underrepresented compared to non-users.

By answering this question, we can gain insights into effectiveness and efficiency, but targeting is only partially informative about them. Regarding the former, targeting does not imply causality in the sense that people who retired under the exemption were able to do so because of it. Regarding the latter, targeting is only a small part of efficiency, as well-targeted but expensive measures can still be inefficient.

The difficulty in answering this question is twofold. Firstly, there is no clear definition of what jobs are classified as "demanding", and different aspects of work can make a job demanding (van den Bogaard et al., 2016). Secondly, even when agreeing on which aspects make a job demanding, there is no data available on these aspects for the full population. Hence, we will leverage Machine Learning to assess targeting along several dimensions of demanding work.

1.2 Procedure and problems

To evaluate the exemption, we will compare the group of exemption users to 1) the employed population, represented by a random sample of individuals who appear in the Dutch employment register, and 2) the exemption-eligible population, represented by a subsample of the employed population with a similar age distribution as exemption users. Since the exemption came into effect in 2021, all users should have been part of the employed population in the year prior, so we evaluate whether users had demanding jobs in 2020.

To account for the absence of population-wide data on demanding jobs, we use a Machine Learning model to predict whether individuals had jobs related to several aspects of demanding work. As predictors, we use individual-level data from the CBS microdata, including, among others, education level, gender, and job-related data such as salary, extra hours worked, and job sector. Since there is no clear, general definition of what jobs are classified as demanding, several aspects of demanding work are considered. In particular, as a proxy for demanding work, we use responses to questions such as "Did/do you work irregular hours?" and "Did/do you need to lift heavy objects?" from the Longitudinal Internet Studies for the Social Sciences (LISS) panel Work & Schooling survey (Scherpenzeel and Das, 2010).

We are primarily interested in the fraction of users who had jobs associated with each aspect of demanding work and in whether they are over- or underrepresented relative to the employed and exemption-eligible populations. Moreover, by evaluating the relationship between exemption use and responses to these questions, we can gain insights into which aspects of demanding work are associated with exemption use and whether the intended audience is reached.

A challenge of this procedure is that the prediction model is unlikely to assign correct labels to all individuals. In other words, our proxies for demanding work are likely to be subject to some degree of misclassification. There are methods that account for this, but to yield correct results, knowledge of misclassification rates is required. Because the exemption users and control groups did not complete the survey, true values are unavailable, and prediction errors cannot be estimated.

First, we perform a benchmark analysis under the assumption that the available estimates are also valid for exemption users and control groups, and investigate the robustness of our results to this assumption. Then, we propose a Bayesian importance sampling procedure that allows relaxation of the nondifferentiality assumption, where we use prior distributions informed by estimates under nondifferentiality. To account for uncertainty about transportability, we assess how results change across different degrees of informativeness of available estimates.

1.3 Structure of this thesis

The remainder of this thesis is structured as follows: Section 2 covers the RVU levy exemption, the *Gezond naar het Pensioen* agreement, and definitions of demanding work more in-depth. Section 3 formalises the problem at hand, gives theoretical prerequisites, and discusses our methodology. Section 4 describes the available data and performs some preliminary analyses. Section 5 discusses the prediction model and its performance. Section 6 uses the generated predictions to evaluate the RVU levy exemption and evaluates how confident we can be in these results. Section 7 concludes.

2 Background

To better understand the situation at hand, this section introduces the RVU levy exemption in more detail, followed by a more thorough description of the proposed changes in the "Gezond naar het Pensioen" agreement. Lastly, we discuss which jobs are classified as demanding, and the assessment framework developed by TNO (TNO, 2025).

2.1 The RVU levy exemption

In 2021, a new regulation introduced the possibility of tax benefits when retiring up to 3 years before the SRA. This regulation, the RVU levy exemption, was specifically intended to help employees with demanding jobs retire early (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a). We will introduce the exemption step-by-step by answering three questions.

What is an RVU?

An early retirement scheme (*Regeling voor Vervroegde Uittreding* (RVU), in Dutch) is an arrangement intended to bridge the financial gap between work and the SRA, in which the employer pays the employee from the time they stop working until the SRA, either at once or periodically. The arrangement can be made either individually between an employer and an employee, or collectively, for example, through a *collectieve arbeidsovereenkomst* (CAO). The conditions for qualification as RVU, judged on objective conditions rather than the motives of either the employer or employee, are twofold: 1) The dismissal has to be age-related, not dysfunction, reorganisation, or similar, and 2) the payment has to be large enough such that it can sustain the employee until retirement, determined by the 70%-toets. More details of the requirements can be found in Ministerie van Sociale Zaken en Werkgelegenheid (2025a).

What is the RVU levy?

In a regular situation, when an RVU is agreed upon, a pseudo final levy has to be paid over the agreed amount. The applicable levy rate was 52% in 2025 but will gradually increase to 65% in 2028. This levy, known as the RVU levy, is meant to discourage people from agreeing on an RVU and keep people in the workforce longer (Ministerie van Sociale Zaken en Werkgelegenheid, 2025a).

What is the RVU levy exemption?

The exemption threshold Early Retirement Scheme (*drempelvrijstelling RVU-heffing*), or RVU levy exemption, is a temporary relaxation of the RVU levy introduced on January 1st, 2021. Here, "threshold exemption" refers to the fact that only the amount below a certain threshold is exempt from the levy. In this case, the levy need not be paid on amounts up to the equivalent of the net AOW amount (€ 2,273 in 2025 and €2,657 in 2026). The RVU levy exemption applies only to the three years preceding the SRA and thus lasts at most 36 months.

2.2 Gezond naar het Pensioen

Originally, the exemption was set to last until the end of 2025, but following the agreement *Gezond naar het Pensioen*, agreed on in 2024 (Ministerie van Sociale Zaken en Werkgelegenheid, 2024) and presented in 2025 (Ministerie van Sociale Zaken en Werkgelegenheid, 2025c), it has been extended from 2026 onwards with the aim to become structural. However, it should become more targeted and involved parties should also focus on sustainable employability. That is, the aim should not only be to support employees who can no longer perform their work, but also to help them transition to gradually less demanding work as they get older. The exemption will be subject to annual reporting and will be evaluated in 2028 and 2031 to determine whether the intended goals are met or a redirection is needed.

During the period 2021-2025, there were no clear guidelines on which jobs were considered demanding and should, hence, be eligible for an RVU. Ministerie van Sociale Zaken en Werkgelegenheid (2025d) shows that this allowed for the presence of generic schemes not targeted at employees with demanding work. Moreover, almost no CAOs demarcated RVU use based on objective criteria. Hence, for the upcoming year, legislators want RVUs to be used more targeted and based on objective criteria.

To this end, an independent third party, TNO, has developed an assessment framework to determine what work is considered demanding in specific sectors. In the coming years, the new "TNO Expertisecentrum Zwaar Werk" will assess whether descriptions of RVU-eligible positions in new collective agreements align with these objective criteria. Based on the criteria discussed in the next subsection, CAO parties must provide documentation for each RVU-eligible function, including a description of the work, which load fields are part of the job, whether they pose a health risk, and whether they are avoidable. Then, TNO will validate this and advise the intended parties on whether the RVU-eligibility aligns with the intended goal. However, this advice is not binding and social partners remain responsible for determining who is eligible for an RVU.

2.3 Demanding jobs

The discussion about which jobs are considered demanding is still ongoing. Bakker and Demerouti (2007) define demanding jobs as "jobs that require sustained physical and/or psychological effort". van den Bogaard et al. (2016) argues that psychologically stressful and psychosocially demanding jobs should also be taken into account, not just physically demanding ones. The Federatie Nederlandse Vakbeweging (FNV) argues there are three types of demanding jobs: 1) Physically demanding jobs that require heavy lifting and repetitive movements, 2) work that requires irregular work hours and night shifts, and 3) mentally demanding work that is emotionally heavy or with verbal abuse (Federatie Nederlandse Vakbeweging, 2024).

In 2025, TNO presented a new assessment framework to more objectively classify which jobs are demanding, specifically designed to validate whether descriptions of RVU-eligible positions in CAOs are intended for the intended users (TNO, 2025). They specify the following criteria: the work must be demanding and pose health risks, the health risks must be inherent to the function and unavoidable, and expertise has been used to determine what functions are demanding and should, thus, be eligible for an RVU.

Additionally, they provide a basic list with aspects that relate to demanding work, spread across 5 load fields:

- Working Times: either during the night, very early or very late, or on-call shifts.
- Psychosocial Load: socially unsafe, emotionally taxing, functional- or social support.
- Cognitive Load: attention, perception capacity, decision-making ability, memory, time pressure
- Physical Load: lifting and carrying, pushing and pulling, unfavourable postures/movements, hand-arm tasks, vibrations and shocks
- Environmental Load: dangerous conditions, biological agents, radiation, heat/cold load, noise

They specifically state that this list is not exhaustive, and that some aspects are satisfied only in combination with others. See TNO (2025) for more details.

To proxy these load fields of demanding work, we use responses to several questions related to these load fields. Table 1 shows the questions used and the load field they relate to.

Label	Question	TNO load field
(a)	"Do/did you work irregular hours?"	Working times
(b)	"My job is/was physically demanding"	Physical load
(c)	"Is/was your job physically demanding?"	Physical load
(d)	"Do/did you need to lift heavy objects?"	Physical load
(e)	"Do/did you need to kneel or stoop?"	Physical load

Table 1: Survey questions considered in this thesis and the load fields they correspond to, as specified by TNO.

As seen in Table 1, the questions considered do not cover all aspects provided by TNO. Hence, we investigate targeting along these dimensions rather than relying on a single measure of demanding work.

3 Theoretical framework

We aim to evaluate whether the RVU levy exemption was used by the intended group: individuals who had demanding jobs. To this end, we assess the fraction of users who had demanding jobs in 2020, before the exemption came into effect, and whether they are over- or underrepresented compared to the employed and exemption-eligible populations.

Our measures for whether people in this group had demanding work are generated by a Machine Learning model that predicts responses to survey questions. As these predictions are subject to misclassification and true values are only available for those who took the survey, misclassification rates cannot be estimated directly. As a result, analyses can only be performed by making assumptions about transportability of these misclassification rates. Assuming they can be computed based on available responses, estimates can be made, and uncertainty can be quantified. However, since this assumption need not hold, we propose a method that allows it to be relaxed.

The remainder of this section formalises the problem at hand, introduces the theoretical prerequisites of our approach, and discusses the methodology. Section 3.1 formalises how we evaluate the exemption's targeting. Sections 3.2 - 3.4 review basic misclassification theory and methods that allow correcting for misclassification when misclassification rates are known. Section 3.5 discusses how predictions can lead to misclassification and how misclassification rates can be estimated on a test set, and Section 3.6 covers the assumptions required for these estimates to be transportable to exemption and control group predictions. Lastly, Section 3.7 proposes a Bayesian importance sampling procedure that allows relaxation of the transportability assumption.

3.1 Entities of interest

Let Y be a binary variable denoting whether someone made use of the RVU levy exemption ($Y = 1$) or not ($Y = 0$), and let X be a binary variable denoting whether someone had a demanding job ($Y = 1$) or not ($Y = 0$). Following the notation of Buonaccorsi (2010), let

$$\alpha_y = P(X = 1 \mid Y = y) = 1 - P(X = 0 \mid Y = y), \tag{1}$$

be the probability that someone with RVU usage $Y = y$ had a demanding job. Moreover, let

$$\Psi = \frac{\alpha_1(1 - \alpha_0)}{\alpha_0(1 - \alpha_1)}, \tag{2}$$

denote the odds ratio, which we use to assess whether people who had demanding jobs are over- or underrepresented among exemption users, compared to non-users. An odds ratio $\Psi > 1$ indicates that demanding jobs are overrepresented, $\alpha_1 > \alpha_0$. The odds ratio is symmetric, in the sense that switching the values of α_1 and α_0 is equivalent to inverting the odds ratio, and it is insensitive to sampling, which makes it a suitable measure in our case-control design.

Alternatively, one might suggest that a regression of the form $Y = \alpha + \beta X + \gamma Z$ would be appropriate to account for the effect of potentially relevant covariates Z, but we argue that the interest lies more in population shares rather than the explanatory power of demanding work.[1]

In our setting, we pre-stratify on exemption use and sample independently from the control groups consisting of individuals who did not use the exemption. This case-control setting simplifies comparisons with similar groups, but it prevents us from investigating the fraction of individuals with demanding work who use the exemption. Formally, this setting allows us to make statements about α, as specified in Table 2, and functions thereof, but it prevents us from doing the same about $P(Y \mid X)$, the probability that someone used the exemption (Buonaccorsi, 2010) given they had a demanding job or not.

3.2 The misclassification problem

The apparent problem is that there is no directly available data on demanding jobs for exemption users. That is, we know the number of people who used the exemption, but we do not know how many of them had demanding work. Hence, to estimate α_1 and α_0, we generate predictions of whether individuals had demanding work, denoted by $\hat{X}$. An overview of counts and conditional probabilities that are important in this research can be found in Table 2.

Imperfections of these predictions imply that our demanding work measures are subject to misclassification, and that our predictions $\hat{X}$ may differ from the true value X. Hence, whether an exemption user actually had demanding work can differ from our prediction, and $P(X \mid Y) = P(\hat{X} \mid Y)$ does not hold trivially. Therefore, naively estimating α_y by

$$\hat{a}_y = \frac{\dot{n}_{1y}}{\dot{n}_{1y} + \dot{n}_{0y}}, \tag{3}$$

where $\dot{n}_{\hat{x}y}$ is the number of individuals with demanding work prediction $\hat{X} = \hat{x}$ and exemption usage $Y = y$, can yield a biased estimate, that either over- or underestimates the fraction of individuals who had demanding work (Buonaccorsi, 2010).

There are several methods that appropriately account for measurement error, where the suitability of each depends on the available data and the problem at hand (see Buonaccorsi (2010) for an introduction and overview). We are interested in the relationship between two binary variables, exemption usage Y and whether an individual had a demanding job X, where our observation $\hat{X}$ may be misclassified. This is referred to as misclassification in two-way tables.

In this setting, correcting for misclassification is straightforward if misclassification probabilities

[1] As an example, imagine a simplified and unrealistic situation where 90% of the individuals in the construction sector have demanding jobs and no individuals outside of the sector do so. Additionally, all individuals in the construction sector use the RVU, whereas none outside do. Then, estimating a regression with sector as a covariate would attribute all explanatory power to the sector and none to the demanding work variable, implying that the regulation does not work efficiently, whereas it reaches all individuals with demanding work and 90% of individuals who use the exemption have demanding work.

	$Y=0$	$Y=1$
$X=0$	n_{00}	n_{01}
$X=1$	n_{10}	n_{11}

(a) n_{xy}

	$Y=0$	$Y=1$
$\hat{X}=0$	$\dot{n}_{00}$	$\dot{n}_{01}$
$\hat{X}=1$	$\dot{n}_{10}$	$\dot{n}_{11}$

(b) $\dot{n}_{\hat{x}y}$

	$X=0$	$X=1$
$\hat{X}=0$	$\ddot{n}_{00y}$	$\ddot{n}_{01y}$
$\hat{X}=1$	$\ddot{n}_{10y}$	$\ddot{n}_{11y}$

(c) $\ddot{n}_{\hat{x}xy}$

	$Y=0$	$Y=1$
$X=0$	$1-\alpha_0$	$1-\alpha_1$
$X=1$	α_0	α_1

(d) $P(X \mid Y)$

	$Y=0$	$Y=1$
$\hat{X}=0$	$1-a_0$	$1-a_1$
$\hat{X}=1$	a_0	a_1

(e) $P(\hat{X} \mid Y)$

	$X=0$	$X=1$
$\hat{X}=0$	$\theta_{0\|0y}$	$\theta_{0\|1y}$
$\hat{X}=1$	$\theta_{1\|0y}$	$\theta_{1\|1y}$

(f) $P(\hat{X} \mid X, Y=y)$

Table 2: An overview of counts and conditional probabilities relating to outcomes Y, true values X, and predictions $\hat{X}$.

are known. Following Buonaccorsi (2010), let

$$\theta_{\hat{x}|x} = P(\hat{X} = \hat{x} \mid X = x) \tag{4}$$

denote the probability of our prediction being $\hat{X} = \hat{x}$, given that the true value is $X = x$. Under misclassification, the probabilities of correct classification $\theta_{1|1}$ and $\theta_{0|0}$, respectively referred to as the sensitivity and specificity, are strictly smaller than one. Equivalently, the misclassification probabilities $\theta_{1|0} = 1 - \theta_{0|0}$ and $\theta_{0|1} = 1 - \theta_{1|1}$ are strictly larger than zero.[2]

3.3 The nondifferentiality problem

It is non-trivial that misclassification probabilities are equivalent between subpopulations. That is, $\theta_{1|1}$ and $\theta_{0|0}$ may differ for different subgroups. For example, let

$$\theta_{\hat{x}|xy} = P(\hat{X} = \hat{x} \mid X = x, Y = y) \tag{5}$$

be the probability of classifying demanding work as $\hat{X} = \hat{x}$ given the true value $X = x$ and exemption use $Y = y$, then it could be that $\theta_{\hat{x}|x1} \neq \theta_{\hat{x}|x0}$ and misclassification rates are differential with respect to Y. Here, they are differential in the sense that they differ between exemption users and non-exemption users.

Opposed to differentiality, nondifferentiality assumes that misclassification rates are equivalent between groups. In the situation of nondifferentiality of misclassification probabilities with respect to exemption usage, we can write

$$P(\hat{X} \mid X, Y) = P(\hat{X} \mid X), \tag{6}$$

[2]As we assume the true X is fixed, we are looking at misclassification. In a setting where the observed is fixed, following Buonaccorsi (2010) notation, the entities of interest are reclassification probabilities $\lambda_{x|\hat{x}} = P(X = x \mid \hat{X} = x)$.

or equivalent $\theta_{\hat{x}|x,y} = \theta_{\hat{x}|x}$.

For the population of survey respondents, the sensitivity and specificity can be adequately estimated by holding out a random test set on which the prediction model is not trained but only tested. However, because the true responses of exemption users and control samples are generally unavailable, we cannot estimate them directly and must make assumptions about the transportability of test set estimates to these predictions.

When the assumption of nondifferentiality of misclassification rates holds, estimates based on the test set are transportable to predictions, in the sense that they are also valid for predictions.

3.4 Estimators

3.4.1 Point estimators

When the true responses are available, following Buonaccorsi (2010), the sensitivity and specificity of predictions can be estimated by

$$\hat{\theta}_{\hat{x}|xy} = \frac{\ddot{n}_{\hat{x}xy}}{\ddot{n}_{\hat{x}xy} + \ddot{n}_{(1-\hat{x})xy}}, \tag{7}$$

where $\ddot{n}_{\hat{x}xy}$ denotes the number of individuals with prediction $\hat{X} = \hat{x}$, true value $X = x$, and exemption use $Y = y$, as in Table 2. Using these estimates, we specify a corrected estimate of α_y as

$$\hat{\alpha}_y = \frac{\hat{a}_y - (1 - \hat{\theta}_{0|0y})}{\hat{\theta}_{1|1y} + \hat{\theta}_{0|0y} - 1}, \tag{8}$$

where $\hat{a}_y$ is the naive estimate of α_y as in Eq. 3, and $\hat{\theta}_{1|1y}$ and $\hat{\theta}_{0|0y}$ denote the estimated sensitivity and specificity, respectively, given $Y = y$, as in Eq. 7. Section 3.6 briefly discusses the intuition behind this formula.

Lastly, using these corrected estimates $\hat{\alpha}_y$, the odds ratio Ψ is estimated by

$$\hat{\Psi} = \frac{\hat{\alpha}_1(1 - \hat{\alpha}_0)}{\hat{\alpha}_0(1 - \hat{\alpha}_1)}. \tag{9}$$

Note that in our situation, $\ddot{n}_{\hat{x}xy}$ is generally unavailable for exemption users and control samples because true responses are not available. Therefore, direct estimates of the sensitivity $\hat{\theta}_{1|1y}$ and specificity $\hat{\theta}_{0|0y}$ are not available and hence corrected estimates $\hat{\alpha}_y$ and $\hat{\Psi}$ cannot be estimated directly.

3.4.2 Variance estimators

Following Buonaccorsi (2010), let

$$\hat{V}(\hat{a}_y) = \frac{\hat{a}_y(1 - \hat{a}_y)}{\dot{n}_{0y} + \dot{n}_{1y}}, \quad \hat{V}(\hat{\theta}_{1|10}) = \frac{\hat{\theta}_{1|10}(1 - \hat{\theta}_{1|10})}{\ddot{n}_{010} + \ddot{n}_{110}}, \quad \hat{V}(\hat{\theta}_{0|00}) = \frac{\hat{\theta}_{0|00}(1 - \hat{\theta}_{0|00})}{\ddot{n}_{000} + \ddot{n}_{100}},$$

denote the binomial variance estimates of $\hat{a}_y$ and $\hat{\theta}_{\hat{x}|xy}$, where $\dot{n}_{\hat{x}y}$ and $\ddot{n}_{\hat{x}xy}$ are defined as in Table 2. Note that we specify the variances of $\hat{\theta}_{\hat{x}|xy}$ only in terms of $Y = 0$, as these can only be estimated on the test set.

Under nondifferential misclassification, estimates of the sensitivity $\hat{\theta}_{1|1y}$ and specificity $\hat{\theta}_{0|0y}$ are transportable to exemption users and control samples. When this is valid, the variances of $\hat{\alpha}_y$ and $\hat{\Psi}$ are estimated using the approximate delta method, which uses the first-order Taylor expansion to approximate the variance of a nonlinear function of an estimator (Buonaccorsi, 2010). Under this assumption, we estimate the variance of the corrected estimate $\hat{\alpha}_y$ by

$$\hat{V}(\hat{\alpha}_y) = \frac{1}{\left(\hat{\theta}_{0|00} + \hat{\theta}_{1|10} - 1\right)^2} \left[\hat{V}(\hat{a}_y) + \hat{\alpha}_y^2 \hat{V}(\hat{\theta}_{1|10}) + (1 - \hat{\alpha}_y)^2 \hat{V}(\hat{\theta}_{0|00})\right]. \tag{10}$$

Note, again, that this specification differs slightly from that of Buonaccorsi (2010), as we estimate the sensitivity $\hat{\theta}_{1|10}$ and specificity $\hat{\theta}_{0|00}$ on the test set.

Estimates for the variance of the odds ratio can be obtained by applying the delta method to compute the variance of the log-odds ratio and then transforming the resulting confidence interval back to the odds ratio by exponentiating. Let $\hat{L}$ denote the logarithm of the estimated odds ratio $\hat{\Psi}$ such that

$$\hat{L} = \log(\hat{\alpha}_1) + \log(1 - \hat{\alpha}_0) - \log(\hat{\alpha}_0) - \log(1 - \hat{\alpha}_1),$$

then, again using the approximate delta method, an estimate of the variance is given by

$$\hat{V}(\hat{L}) = \frac{\hat{V}(\hat{\alpha}_0)}{\hat{\alpha}_0^2(1 - \hat{\alpha}_0)^2} + \frac{\hat{V}(\hat{\alpha}_1)}{\hat{\alpha}_1^2(1 - \hat{\alpha}_1)^2} - 2\frac{\widehat{cov}(\hat{\alpha}_0, \hat{\alpha}_1)}{\hat{\alpha}_0(1 - \hat{\alpha}_0)\hat{\alpha}_1(1 - \hat{\alpha}_1)}, \tag{11}$$

where, since $\hat{\alpha}_0$ and $\hat{\alpha}_1$ depend on the same estimates for the sensitivity and specificity, the covariance is nonzero and can be estimated by

$$\widehat{cov}(\hat{\alpha}_0, \hat{\alpha}_1) = \frac{1}{\left(\hat{\theta}_{0|00} + \hat{\theta}_{1|10} - 1\right)^2} \left[\hat{\alpha}_0 \hat{\alpha}_1 \hat{V}(\hat{\theta}_{1|10}) + (1 - \hat{\alpha}_0)(1 - \hat{\alpha}_1) \hat{V}(\hat{\theta}_{0|00})\right].$$

A confidence interval for L is then given by $\hat{L} \pm z_{\frac{a}{2}} \sqrt{\hat{V}(\hat{L})} = [a, b]$, which is translated to a confidence interval for Ψ by exponentiating, $\left[e^a, e^b\right]$. A known limitation of delta-method variance estimates is that the imposed linearity can produce inadmissible estimates outside the parameter space.

3.5 Misclassification due to prediction errors

We predict whether individuals have or had demanding work using a Machine Learning model. Let $f(\cdot)$ denote the model and let Z denote the features used as predictors. Then predicted values $\hat{X} = \hat{f}(Z)$ are a deterministic function of these features, and training the prediction model amounts to learning about the conditional distribution $P(X \mid Z)$ to determine which regions of Z should map to $\hat{X} = 0$ and which to $\hat{X} = 1$. That is, training $f(\cdot)$ comes down to determining $\hat{A}_0$ and $\hat{A}_1$ such that

$$\hat{A}_{\hat{x}} = \left\{ z \in \mathcal{Z} : \hat{f}(z) = \hat{x} \ \forall \ z \in \hat{A}_{\hat{x}} \right\}, \quad \hat{x} \in \{0, 1\}.$$

Since our model should predict well for both demanding and non-demanding work, simply using the prediction accuracy $P(\hat{X} = X)$ as a metric would not work well if class imbalance is present. Rather, we look at the conditional distribution $P(Z \mid X)$ and aim to approximate an optimal $f^*(\cdot)$ that gives A_0^* and A_1^*, where

$$A_x^* = \{ z \in \mathcal{Z} : P(z \mid X = x) > P(z \mid X = 1 - x) \}, \quad x \in \{0, 1\}, \tag{12}$$

which maximises the mean of the sensitivity and specificity. This is attained by setting $f(z) = \hat{x}$, where $X = \hat{x}$ is the group with the highest density at $Z = z$. A simplified visualisation of this can be found in Figure 1.

Given the trained predictor, predictions for new samples are made by predicting $\hat{X} = 1$ if $z \in \hat{A}_1$, and $\hat{X} = 0$ otherwise. To avoid overestimating prediction performance due to overfitting, it is important to evaluate performance on an independent test set. The simplified visualisation in Figure 2 shows how misclassification probabilities θ can be estimated by calculating the fraction of each group that appears in $\hat{A}_0$ and $\hat{A}_1$, respectively.

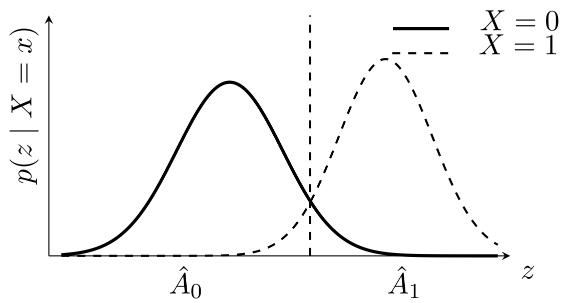


Figure 1: Distributions of features Z conditional on X. Training the predictor yields $\hat{A}_0$ and $\hat{A}_1$, where $f(z)$ maps to the x with the highest conditional density $P(Z = z \mid X = x)$.

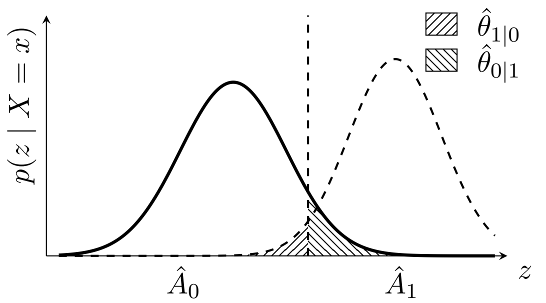


Figure 2: Evaluating performance on test set gives $\hat{\theta}_{(1-x)|x}$, the area under conditional density $P(Z \mid X = x)$ that is in $\hat{A}_{(1-x)}$. The imperfection of the decision boundary is due to sampling, approximation, and estimation errors.

In this process, misclassification can occur due to several reasons. Unless the Z-regions of $X = 0$ and $X = 1$ are completely separated, even the optimal prediction model $f^*(\cdot)$ produces misclassified predictions. Here, misclassification probabilities $\theta_{(1-x)|x}$ depend on the fraction of the conditional density $P(Z \mid X = x)$ in the area $\hat{A}_{(1-x)}$, as shown in Figure 2. Formally,

$$\theta_{(1-x)|x} = \int_{z \in \mathcal{Z}} \mathbb{1}_{\{z \in \hat{A}_{(1-x)}\}} P(Z = z \mid X = x) = \int_{z \in \hat{A}_{(1-x)}} P(Z = z \mid X = x), \quad x \in \{0, 1\}. \tag{13}$$

Furthermore, the scenario where our trained model $\hat{f}(\cdot)$ finds the optimal A_0^* and A_1^* is unlikely. Because we do not have infinite data, the limited training data will not contain sufficient information to learn the optimal regions, leading to estimation error. Additionally, due to sampling error, the training sample will not be a perfect representation of the full population. Lastly, some models suffer from approximation error due to restrictions on their functional form, which prevent them from forming optimal regions even with infinite data. However, this is more of a problem for traditional econometric models rather than recent, more flexible ML models. The effects resulting from suboptimal decision regions can also be seen in Figure 2.

Looking at Figure 2, it is easily explained why the formula for the corrected estimate $\hat{\alpha}_y$ in Eq. 8 makes sense intuitively. Using $P(\hat{X} = 1 \mid X = 1) = \theta_{1|1}$ and $P(\hat{X} = 0 \mid X = 0) = \theta_{0|0}$, the probability that an individual will be classified as $\hat{X} = 1$ is equal to

$$P(\hat{X} = 1) = \theta_{1|1} P(X = 1) + (1 - \theta_{0|0}) P(X = 0), \tag{14}$$

which is a very intuitive formula. Rewriting this equation gives

$$P(X = 1) = \frac{P(\hat{X} = 1) - (1 - \theta_{0|0})}{\theta_{1|1} + \theta_{0|0} - 1},$$

which corresponds to the corrected estimator in Eq. 8. As this explanation again shows, accurate estimates of the misclassification rates $\theta_{0|0}$ and $\theta_{1|1}$ are required for valid inference.

3.6 Nondifferentiality of prediction errors

For subsamples with known true values, the prediction performance can be accurately estimated by comparing predictions against these true values. Additionally, performance can be estimated on a subgroup of the test set, such as a specific age group, where differences in the conditional distribution $P(Z|X)$ can occur. Figure 3 illustrates a simplified example of how prediction errors can differ between subgroups and how they can be estimated.

When making predictions for samples whose true values are unknown, estimating misclassification rates directly is not possible. As Figure 4 demonstrates, the fraction of individuals in $\hat{A}_0$ and $\hat{A}_1$ for each category is known, but estimating misclassification rates of the predictions is not feasible directly, but requires assumptions. A straightforward, but perhaps unlikely, assumption would be nondifferentiality of prediction errors.

Combining Eq. 6 and Eq. 13, nondifferentiality of prediction errors between $Y = 0$ and $Y = 1$

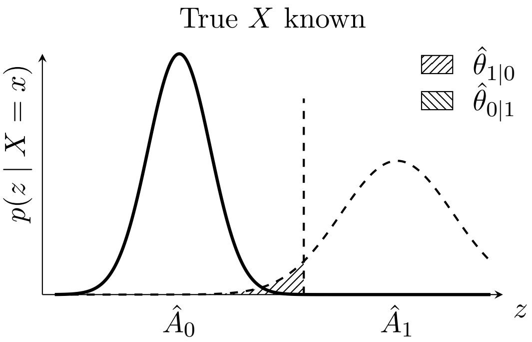


Figure 3: A simplified visualisation of how prediction performance can be evaluated for a subgroup. This yields an alternative $\hat{\theta}$.

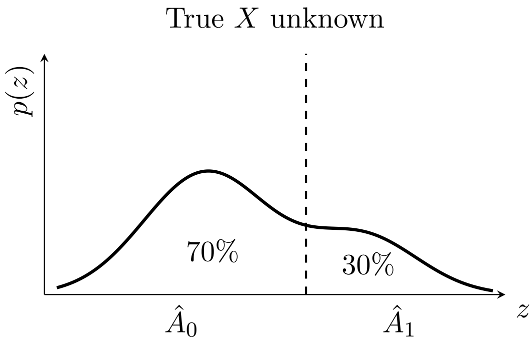


Figure 4: When true X is unknown, prediction rates can be computed, but misclassification rates cannot be estimated.

requires

$$\int_{z \in A_{1-i}} P(Z = z \mid X = x, Y = 0) = \int_{z \in A_{1-i}} P(Z = z \mid X = x, Y = 1), \quad x \in \{0, 1\}, \tag{15}$$

which implies the density of Z given $X = x$ must have the same weight in regions $\hat{A}_{\hat{x}}$ for $Y = 0$ compared to $Y = 1$. Intuitively, this means that in Figures 2 and 3 the shaded region under the conditional distributions $P(Z \mid X = x)$ must be of equal area between figures. This can be true in two ways: i) $P(Z \mid X, Y) = P(Z \mid X)$, they have the same distribution, or ii) $P(Z \mid X, Y) \neq P(Z \mid X)$, but the weight is distributed such that Eq. 15 still holds.

We argue that the former could hold structurally under restrictive assumptions, whereas the latter is very unlikely to hold structurally. Namely, i) implies that differences in $P(\hat{X} \mid Y = y)$ between $Y = 0$ and $Y = 1$ are only due to differences in $P(X \mid Y)$, as it cannot be due to differences in $P(Z \mid X, Y)$. This would be valid if, given demanding work, exemption usage is a random sample of individuals. Regarding ii), we hypothesize there is no simple sampling procedure that would satisfy this, so we argue it could hold only to a reasonable extent by chance.

Nevertheless, we argue too that regarding exemption users, ii) is also unlikely to hold, as we hypothesise that not all individuals with demanding jobs have the same probability of using the exemption. For example, if RVU users are more concentrated in specific sectors, $P(Z \mid X, Y)$ will have a different distribution between $Y = 0$ and $Y = 1$, and hence nondifferentiality as in Eq. 15 is unlikely to hold. Regarding control groups it could be valid, as they are meant to be a representative sample of their respective populations, as argued in Section 4.

We can, however, be confident in ruling out certain combinations of sensitivity and specificity, as some combinations would be highly unlikely to yield the observed prediction ratios. For example, in Figure 4, a specificity of 50% and a sensitivity of 100%, following Eq. 14, would give

$$P(\hat{X} = 1) = P(X = 1) + 0.5P(X = 0),$$

which would yield $P(\hat{X}) \in [0.5, 1]$ for al feasible combinations of $P(X)$. Hence, we argue that this combination of sensitivity and specificity would be highly unlikely to yield 30% of observations with $\hat{X} = 1$, especially for a large number of observations. This idea of using the likelihood of parameter combinations given the observed data is the foundation of the method proposed in Section 3.7.

Concluding, we argue that the assumption of transportability of misclassification rates from the test set to control samples could hold, but transportability to exemption users is unlikely to be valid because we expect that exemption users are not a random sample of the populations with and without demanding work. However, we hypothesise that these estimates can still be informative of the true sensitivity and specificity of predictions, which Section 3.7 leverages.

3.7 Bayesian importance sampling

Let $\Theta_p = \{\theta \in \Theta : 0 < \alpha_1, \theta_{0|0}, \theta_{1|1} < 1,\ \theta_{0|0} + \theta_{1|1} > 1\}$ denote the space spanning all possible parameter combinations, where the former constraint holds trivially as all probabilities are nonzero, and the latter implies that our model has some predictive power. Also, let

$$p_\theta = P(\hat{X}_i = 1 \mid \theta) = \theta_{1|1}\alpha + (1 - \theta_{0|0})(1 - \alpha), \quad i \in \{1, \ldots, n\} \tag{16}$$

denote the probability of observing a success ($\hat{X} = 1$) given $\theta = (\alpha, \theta_{0|0}, \theta_{1|1})$. Furthermore, let k denote the number of observed successes, and let n be the total number of observations, such that $a = k/n$. The probability of observing a fraction of a successes given parameters θ and n draws then equals

$$\begin{aligned} P(a \mid \theta, n) &= P(k \mid \theta, n) \\ &= \binom{n}{k} p_\theta^k (1 - p_\theta)^{(n-k)}, \end{aligned} \tag{17}$$

the (scaled) binomial pmf with p_θ specified as in Eq 16.

Given θ, and following the law of large numbers, the distribution of a gets tighter around its mean p_θ and as n increases. Hence, as n increases, observations further from p_θ become less likely. In this case, the distribution in Eq. 17 no longer specifies a distribution, but rather a point with all probability mass. As a result, p_θ in Eq. 16 must match the observed fraction of successes a, restricting the possible combinations of α, θ_0 and θ_1 significantly, losing one degree of freedom.

Using Bayes' rule, the posterior probability of θ given observations equals

$$P(\theta \mid a, n) = \frac{P(a \mid \theta, n) P(\theta, n)}{P(a, n)},$$

which specifies a distribution for the true parameter θ given prior knowledge $P(\theta, n) = p(\theta)$ and $P(a, n) = P(a \mid n)$, treating n fixed (note that prior distribution $P(\theta)$ is not the same as probability p_θ). Since $P(a \mid \theta, n)$ converges to p_θ as $n \to \infty$, again the likely combinations of α, θ_0 and θ_1 converge to just the combinations that equate p_θ as in Eq. 16 to the observed fraction of successes a.

Using this specification, the posterior probability of θ lying in a specific region Θ_0 is given by

$$\begin{aligned} P(\theta \in \Theta_0 \mid a) &= \int_{\Theta_0} P(\theta \mid a)\, d\theta \\ &= \frac{\int_{\Theta_0} P(a \mid \theta) P(\theta)\, d\theta}{\int_{\Theta_p} P(a \mid \theta) P(\theta)\, d\theta}, \end{aligned}$$

where the $P(a \mid n)$ term is dropped as it appears both in the numerator and the denominator. Letting $\Theta_1 = \{\theta \in \Theta_p : \alpha_1 > \alpha_0\}$ denote the area for which $\Psi > 1$, and $P(a \mid \theta, n)$ as in Eq. 17, we have

$$P(\Psi > 1 \mid a, n) = \frac{\int_{\Theta_1} p_{1,\theta}^{k_1} (1 - p_{1,\theta})^{(n_1 - k_1)} p_{0,\theta}^{k_0} (1 - p_{0,\theta})^{(n_0 - k_0)} P(\theta_1, \theta_0)\, d\theta}{\int_{\Theta_p} p_{1,\theta}^{k_1} (1 - p_{1,\theta})^{(n_1 - k_1)} p_{0,\theta}^{k_0} (1 - p_{0,\theta})^{(n_0 - k_0)} P(\theta_1, \theta_0)\, d\theta}, \tag{18}$$

where $p_{y,\theta}, n_y, k_y$, and θ_y refer to the parameters for the group $Y = y$, which is the posterior probability of the true parameters satisfying $\alpha_1 > \alpha_0$, given prior distribution $P(\theta_1, \theta_0)$, and observing a given n observations.

For the prior distribution $P(\theta_1, \theta_0)$, we assume independence between θ_1 and θ_0 such that $P(\theta_1, \theta_0) = P(\theta_1)P(\theta_0)$. We can determine the level of informativeness, representing how certain we are about our knowledge regarding the true θ_y, where a more informative prior specifies a more tightly centred distribution, and a less informative prior specifies a wider distribution. For example, we can specify an uninformative prior

$$P(\theta_y) = c, \quad \theta_y \in \Theta_p,$$

a uniform distribution over Θ_p, representing no prior knowledge about what combination of $\alpha_y, \theta_{0|0y}, \theta_{1|1y}$ is more likely. Alternatively, we can specify an informative prior

$$P(\theta_y \mid \bar{\theta}_y, \sigma_{\theta_y}) = \phi(\theta_y \mid \bar{\theta}_y, \sigma_{\theta_y})$$

where $\phi(\cdot)$ is some functional specification, $\bar{\theta}_y$ represents the prior expectation about the true θ_y, and σ_{θ_y} represents the confidence in this expectation, with a higher σ_{θ_y} corresponding to less confidence. For example, assuming a truncated independent Gaussian prior

$$P(\theta_y \mid \bar{\theta}_y, \sigma_{\theta_y}) = \begin{cases} \exp\left(-\frac{1}{2}\left[\frac{(\alpha_y - \bar{\alpha}_y)^2}{\sigma_{\alpha_y}^2} + \frac{(\theta_{0|0y} - \bar{\theta}_{0|0y})^2}{\sigma_{\theta_{0|0y}}^2} + \frac{(\theta_{1|1y} - \bar{\theta}_{1|1y})^2}{\sigma_{\theta_{1|1y}}^2}\right]\right), & \theta \in \Theta_p, \\ 0, & \text{otherwise}, \end{cases}$$

where $\sigma_{\theta_y} = (\sigma_{\alpha_y}, \sigma_{\theta_{0|0y}}, \sigma_{\theta_{1|1y}})$ denotes the standard deviation of each univariate prior.

Under the assumption of a specific prior distributions $P(\theta_0)$ and $P(\theta_1)$, we can estimate the posterior probability in Eq. 18 by importance sampling, where we independently draw parameters from the prior distributions $P(\theta_y)$, and compute the likelihood of observing a_y given those parameters. The posterior probability for $P(\Psi > 1)$ is then given by the weighted fraction of parameter draws such that $\alpha_1 > \alpha_0$, with weights equal to the corresponding likelihood.

Where the estimators in Section 3.4 require transportability of the estimated misclassification rates from the test set to predictions, this importance sampling procedure allows us to use these estimates, but let the uncertainty regarding transportability be reflected in the informativeness of the priors.

4 Data

This thesis uses three data sources: 1) a list of individuals who made use of an RVU in the period 2021-2024, 2) responses to the LISS panel Work & Schooling survey, and 3) administrative data from the CBS microdata. All datasets were uploaded to/accessed in the CBS Microdata Remote Access (RA) environment.

4.1 CBS Microdata

The CBS microdata contains many detailed, anonymised individual-level datasets about the Netherlands. This ranges from personal records such as birth months and salaries to data on households and organisations. Use of this data is subject to strict confidentiality and is available only to approved research institutions.

In this research, we make use of the GBAPERSOONKTAB (Centraal Bureau voor Statistiek, 2021), SPOLISBUS (Centraal Bureau voor Statistiek, 2025), and HOOGSTEOPLTAB (Centraal Bureau voor Statistiek, 2019) microdatasets. The GBA (gemeentelijke basisadministratie) contains information on gender and birth year, the SPOLISBUS contains monthly data from the "polisadministratie" which covers, among others, salary, hours worked, whether the individual had a company car, sector, and extra hours worked. Lastly, the HOOGSTEOPLTAB contains the highest level of education achieved, which is classified as 1 (low), 2 (secondary), or 3 (high) (Centraal Bureau voor Statistiek, 2026).

4.2 RVU users

To define who used the exemption, we use a list of individuals who used an RVU in the period 2021-2024. These individuals appear with code 53 (early retirement scheme) as a type of employment relationship in the SPOLISBUS. Since this data is not directly available in the regular SPOLISBUS, it was provided to us by the UWV. This dataset contains 27,016 individuals who used an RVU in the period 2021-2024.

Appearance in this dataset denotes the use of an RVU. Since there were no hard requirements for using the exemption once an RVU had been reached, and the exemption was also valid for the first part of the sums paid if the amount exceeded the threshold, we argue that appearance in this dataset is a good measure of exemption use. Additionally, the year the RVU was taken is not available, only whether it was before or after January 2021. Hence, the predictions made for 2020 do not represent year-before-retirement predictions but rather pre-exemption-era predictions.

4.3 LISS Work and Schooling survey

As proxies for demanding work, we make use of data from the LISS panel (Longitudinal Internet studies for the Social Sciences), which is administered and managed by the non-profit research institute Centerdata (Tilburg University, the Netherlands) (Scherpenzeel and Das, 2010). Particularly, we use responses to several questions from the LISS Work & Schooling survey, which consists of 17 yearly waves (2008 - 2024). According to the LISS panel, their surveys are representative samples of the Dutch population (Scherpenzeel and Das, 2010), which is investigated in Section 4.4

From the survey, we identified the following five questions as most relevant regarding demanding work:

(a) **Irregular hours**: Do/did you work irregular hours?

(b) **Physically demanding (agree)**: My job is/was physically demanding.

(c) **Physically demanding (often)**: Is/was your job physically demanding?

(d) **Heavy lifting**: Do/did you need to lift heavy objects?

(e) **Kneel / stoop**: Do/did you need to kneel or stoop?

where questions (a), (c), (d), and (e) were answered with never/sometimes/often, and question (b) with strongly disagree/disagree/agree/strongly agree. We convert responses to these questions into a binary indicator X, where $X = 1$ indicates that work is demanding. Answers are classified as demanding if the response is often or agree/strongly agree.

It is important to note that this survey is based on self-reported data and is therefore likely to contain measurement errors. However, we do not take this into account in the analysis. Out of 99,404 responses across different waves, 13,896 belonged to individuals who did not give permission to connect to their responses to the microdata, 1,768 to individuals who could not be found in the microdata, and 83,740 responses were connected. Responses correspond to $13,059$ unique respondents, where the number of participations per respondent ranges from 1 to 17.

4.4 Comparison groups

To evaluate the RVU levy exemption, we train a prediction model using survey responses to compare the fraction of exemption users with demanding jobs to that of the employed population and the exemption-eligible population. Therefore, we construct four groups of employed individuals:

(1) **Respondents**: respondents to the LISS Work & Schooling survey, whose responses are used to train the prediction model.

(2) **Exempted**: people who used the RVU levy exemption.

(3) **Employed**: a random sample of people who appear in the Dutch employment register in 2020 and did not use the exemption.

(4) **Eligible**: a random sample of people who appear in the Dutch employment register in 2020, did not use the exemption, and have a similar age distribution to the exempted group.

From these, true responses are only available for the respondents (1) group. Hence, the prediction model will be trained and evaluated on this group, and predictions will be made with the trained model for the exempted (2), employed (3) and eligible (4) prediction groups. Moreover, the exempted (2) group is our primary interest and the employed (3) and eligibe (4) are control groups.

The respondents (1) and exempted (2) groups are predefined, as we can not influence who participated in the LISS survey or who used the exemption. A small sample of individuals who appear in the LISS survey are also exemption users. As they are assumed to be a random sample of the exempted (2) group, but not a random sample of the respondents (1) group, they are assigned to the respondents (1) group to prevent selection bias.

Since the exemption can only be used by individuals in the employed population, the employed (3) and eligible (4) groups are constructed by randomly selecting individuals from the GBA who appeared in the SPOLIS 2020. For the eligible (4) group, a weighted draw is performed, with weights proportional to the number of exemption users with the same birth year. With "eligible" we refer to the fact that they could have potentially used the exemption, even though they are not necessarily considered to be part of the intended recipients.

Table 3 shows the original sizes of each group and the number of individuals that appear in the 2010-2025 and 2020 editions of the SPOLIS, respectively. Remarkable is the number of individuals in the exempted (2) group, as apparently not all individuals who used an RVU appeared in the SPOLIS between 2010 and 2025, and a larger proportion did not appear in the 2020 SPOLIS.

Inference validity depends on whether misclassification rates estimated for respondents are

Group	n	n_{SPOLIS}	$n_{\text{SPOLIS,2020}}$
(1)-(4)	200,055	129,019	93,362
(1) Respondents	13,059	10,441	7,745
(2) Exempted	27,016	27,006	26,618
(3) Employed	100,000	51,969	31,483
(4) Eligible	60,000	39,623	27,535

Table 3: Number of individuals that appear in the Dutch employment register, by comparison group. n denotes the number of people who appear in the GBA. n_{SPOLIS} and $n_{\text{SPOLIS,2020}}$ denote the number of people that appear at least once between 2010 and 2025, and in 2020, respectively.

transportable to the predictions for the exempted (2), employed (3) and eligible (4) groups. As argued in Section 3.6, this transportability depends on the distribution $P(Z \mid X, Y)$ of features conditional on the subpopulations and whether they had demanding work.

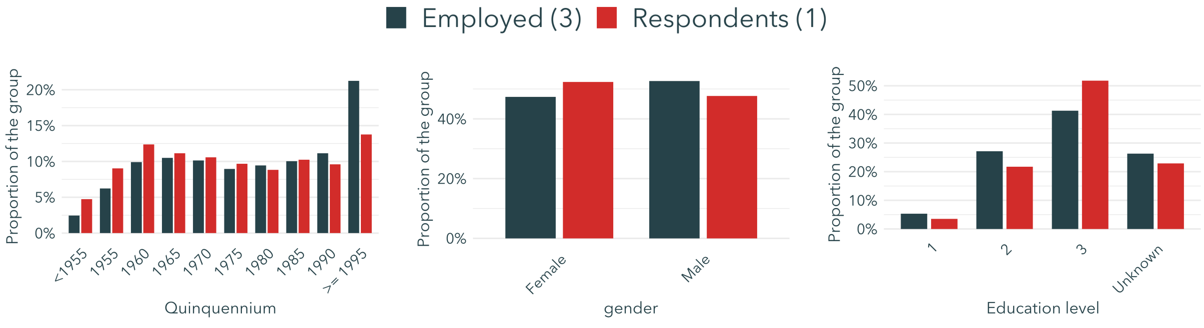
According to the LISS panel, their respondents are a representative sample of the Dutch population. Hence, selecting only respondents who appear in the administrative records for a given age should yield a representative sample of that group in the year of the survey. Additionally, selecting a random sample from the GBA and filtering it using the same criteria should yield a representative sample of the population in that year as well. Therefore, if the LISS respondents are indeed a representative sample of the Dutch population, survey respondents (1) and the employed (3) and eligible (4) control groups should be drawn independently from the same population. As a result, misclassification probabilities estimated on the test set should be transportable to the comparison sets, and valid inference could be performed.

Figures 5 and 6 show the distributions of several characteristics for the comparison groups.[3] For comparison, survey respondents are filtered for appearance in the 2020 employment register, but to increase the sample size while training the ML model, individuals are kept if they appear at least once in the employment register between 2010 and 2025.

Figure 5 shows the distribution of certain characteristics for the respondents (1) and employed (3) groups. The youngest age cohorts are more frequent in the random sample, whereas the oldest are more frequent in the respondents' sample. Also, females are slightly overrepresented in the LISS survey, as are individuals with higher levels of education. These figures suggest that the respondents group (1) is not a representative sample of the employed population (3).

A possible explanation for these differences is that the employed respondents were a random sample of the employed population in the year the survey was taken, but since some respondents took the survey in earlier years, filtering them for appearance in the 2020 SPOLIS does not represent a random sample. Hence, some people are older in 2020 than when they took the

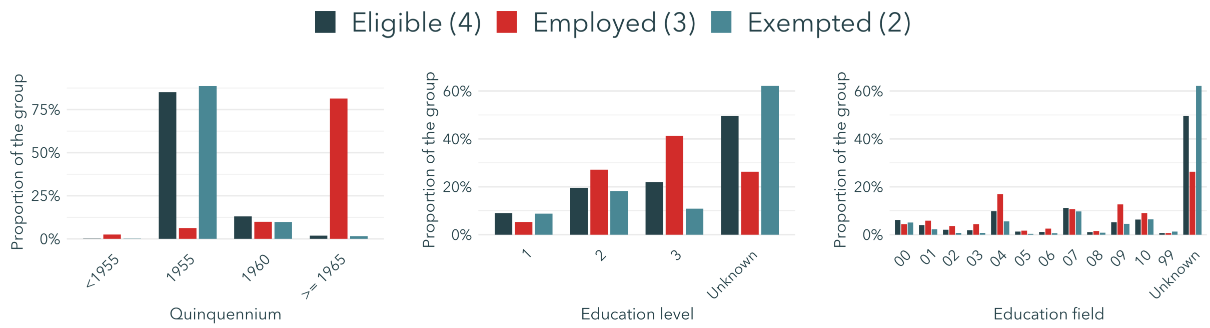
[3]To guarantee privacy, raw datasets can only be accessed in the secure RA environment, and results have to satisfy strict anti-disclosure criteria before they can be exported from the secure environment. One of these criteria is that cells in tables must not be based on fewer than 10 individuals. Therefore, some results report merged groups to comply with these standards.



(a) Birth year (b) Gender (c) Education level

Figure 5: Characteristics distributions of respondents (1) and employed (3) groups.

survey, but still appear in the employment registration, allowing older cohorts to participate in more surveys. Especially the difference for the youngest group would make sense, as someone who was 16 in 2020 could have appeared at most once, whereas someone who was 40 could have appeared at most 17 times. This effect would decay with age, as shown in Figure 5.



(a) Birth year (b) Education level (c) Education field

Figure 6: Characteristics distributions of the exempted (2), employed (3), and eligible (4) comparison groups.

Figure 6 shows the distributions of several characteristics of the exempted (2), employed (3), and eligible (4) comparison groups. It is apparent that there are large differences between the employed group (3), the exempted (2), and the eligible (4) groups. This makes sense as groups (2) and (4) consist of similar-aged individuals, whereas group (3) is not selected based on age. Additionally, differences in education level and education field between groups (2) and (4) suggest that exemption users are not a representative sample of the exemption-eligible population, which aligns with our expectations. Particularly interesting is the large fraction of exemption users for whom no information about education is available.

The results suggest that neither the respondents (1) compared to employed (3), nor the eligible (4) compared to exempted (2) groups have a similar feature distribution, and hence we argue it is unlikely for the assumption of nondifferentiality to hold in practice. However, as respondents are to some extent a random sample of the employed population, we are confident that the estimated misclassification rates are informative of the misclassification rates for the employed (3) and eligible (4) control groups. For the exempted group, we expect misclassification rates to be less informative.

4.5 Implementation limitations

During the construction of the employed (3) and eligible (4) control samples in the CBS RA environment, several implementation errors were made. These include stratification by gender, selecting subsamples in reverse order, and filtering for overlap between the comparison sets. Also, in the weighting procedure, our method overrepresents birth years that appear more frequently in the main sample.

These issues may affect the representativeness of the control samples and, in turn, the transportability of misclassification rates. The direction of the bias imposed by these limitations is not trivially determined and contributes to the uncertainty in the validity of transportability. Moreover, if gender is correlated with demanding work, we hypothesise that our results attenuate differences in prevalence between exemption users and control samples.

Due to the severe computational time required (multiple days) and the requirement to provide the export files before access to the RA environment was revoked, these could not be resolved during writing. We consider this a limitation of our study.

5 Predictions

We use Machine Learning to predict responses to survey questions related to several aspects of demanding work. We train and evaluate two models on the respondents group (1) for which the true answer is known. Using the trained models, we make predictions for the individuals in the exempted (2), employed (3), and eligible (4) groups for the most recent pre-exemption year, 2020. See Section 2 for a list of the questions used and what load field of demanding work they correspond to.

In this procedure, it is important to note that sheer prediction performance is not the primary criterion. As discussed in Sections 1 and 3, an accurate estimate of the prediction performance, specifically the sensitivity and specificity, is more important, as the proposed methodology can account for misclassification when misclassification rates are known.

5.1 Machine Learning models

We use random forest and XGBoost as machine learning models. The main advantage of these models compared to more traditional econometric models is their greater flexibility, as they do not impose a linear relationship between the features and the outcome variable. As we expect the true distribution $P(X \mid Z)$ to be complex and nonlinear, we hypothesise that these models lead to less biased estimates across regions of Z, as they are less susceptible to approximation error.

A common downside of using flexible Machine Learning models is that they provide little insight into the underlying relationships and are, to some extent, a "black box". However, we are only interested in prediction performance, not in the relationship between the features Z and the outcome variable X, so both random forest and XGBoost are suitable models to generate predictions.

5.2 Features

We construct features from the CBS microdata. As both models are flexible and can generally select important variables themselves in a high-dimensional setting (Chen and Guestrin, 2016; Genuer et al., 2008), we use many features. In addition to the features already available in the microdata, we construct additional ones ourselves.

From the monthly administrative SPOLIS dataset, we select only the first month in which each individual appears. To account for differences within each year, new features are constructed, including the variation of salary within the year, the variation in hours worked within the year, and the number of times each person appears within the year. Additionally, we use information about the individuals, such as gender, highest achieved education level and corresponding education field, and age.

As data from many years is available, both in the microdata and the LISS survey, a target year is chosen for which the outcome is predicted. SPOLIS features are taken at several time windows relative to the target year (e.g., 10 years before, 1 year before, the year of), and age is computed for the target year.

For the group of respondents (1), on whom the model is trained, the target year is either the year the survey was completed if the person was working at the time, or the last year the person worked if they were no longer working. For the exempted (2), employed (3), and eligible (4) groups, for whom prediction are made, the target year is set to 2020.

5.3 Training and tuning

As true responses X are available only for LISS respondents, the model is trained and evaluated on group (1). To this end, individuals in this group are split into three sets: 52.5% training, 17.5% validation, 30% test. The test set is relatively large, as the emphasis is on obtaining accurate performance estimates. Here, the split is based on individuals rather than on responses to ensure that no spillover effects occur. Prediction performance is evaluated only once on the test set to avoid overfitting and to produce accurate estimates.

Both models produce a score between 0 and 1, where a value closer to 1 indicates the true value is more likely to be 1, and the score is converted to binary based on whether it exceeds a threshold. Increasing this threshold will lead to fewer individuals being classified as 1, thereby increasing specificity and decreasing sensitivity. Reducing the threshold has the opposite effect.

To align our training procedure with finding the optimal $f^*(Z)$ as in Eq. 12, the threshold is chosen that maximises Youden's J statistic,

$$J = \text{sensitivity} + \text{specificity} - 1,$$

which is equivalent to maximising the average of the sensitivity and specificity.

Additionally, during training, observations are assigned weights equal to the inverse relative frequency, so that groups with and without demanding jobs are treated equally. Parameters are optimised via a grid search based on J and evaluated on the independent validation set. The best model, by J, per question is chosen to generate the labels for individuals in groups (2) - (4), for whom the true value is unknown.

As mentioned in Section 4, some individuals have provided more responses than others. However, a preliminary analysis indicated that responses often changed over time. Also, because the survey responses are self-reported, repeated measures provide greater certainty about the answer. Hence, each response is given equal weight.

5.4 Estimating the sensitivity and specificity

To estimate misclassification rates, we consider predictions on the test set, which consists of $14,477$ responses to each question. Table 4 contains the confusion matrices of the best prediction model for each question.

Table 4 shows that given the true value, our model is more often correct than incorrect. Hence, it has significant predictive power. Due to class imbalance, however, oftentimes more than half of the predictions that someone had demanding work are wrong. This is a limitation of using sensitivity and specificity as metrics, but in our methodology, it is not an issue.

Based on these confusion matrices, the sensitivity and specificity can be estimated using the estimator in Eq. 7. We generate two estimates of the sensitivity and specificity, one based on the entire test set, $\theta_{\hat{x}|x,\text{all}}$, and one based on an age-weighted sample of the test set, $\theta_{\hat{x}|x,\text{age}}$. For the latter, we compute weighted estimates, assigning each individual a weight equal to the fraction of exemption users with the same birth year, thereby mimicking the procedure used to select individuals for the group. The estimated specificities and sensitivities are reported in Table 5.

Table 5 shows that all estimates of the sensitivity and specificity lie between 0.662 and 0.824. This suggests that, in line with our observations in Table 4, our model has significant predictive

X	$\hat{X}$ 0	1
0	8579	3334
1	867	1697

(a) Irregular hours

X	$\hat{X}$ 0	1
0	6939	3769
1	1037	2732

(b) Physically demanding (agree)

X	$\hat{X}$ 0	1
0	8585	4243
1	384	1265

(c) Physically demanding (often)

X	$\hat{X}$ 0	1
0	9390	3884
1	363	840

(d) Heavy lifting

X	$\hat{X}$ 0	1
0	8082	3945
1	616	1834

(e) Kneel / stoop

Table 4: Confusion matrices for predictions of questions (a) - (e) on the test set. X denotes the true value and $\hat{X}$ denotes predictions. A value of one indicates that the predicted answer is "often" to questions (a), (c), (d), and (e), or "agree" or "strongly agree" to question (b).

Q	$\hat{\theta}_{1\|1,\text{all}}$	$\hat{\theta}_{0\|0,\text{all}}$	$\hat{\theta}_{1\|1,\text{age}}$	$\hat{\theta}_{0\|0,\text{age}}$
(a)	0.662	0.720	0.670	0.755
(b)	0.725	0.648	0.706	0.633
(c)	0.767	0.669	0.824	0.710
(d)	0.698	0.707	0.712	0.755
(e)	0.749	0.672	0.750	0.685

Table 5: Point estimates of the unweighted, $\theta_{\hat{x}|x,\text{all}}$, and age-weighted, $\theta_{\hat{x}|x,\text{age}}$, sensitivity and specificity, for each question (a) - (e).

power both for people with demanding jobs and those without. Additionally, it shows that the estimated sensitivities and specificities indeed differ depending on the group on which they are estimated. For question (b), the estimated sensitivity and specificity are lower when evaluated on the age-stratified sample. For the other questions, they are higher.

As mentioned in 4, we argue that we have little confidence that these misclassification rates are transportable to the prediction groups (2) - (4). However, we hypothesise that the unweighted estimate may be informative about the misclassification rates of the employed group (3) and that the age-weighted estimate may be informative about the eligible comparison group (4). Moreover, for the exempted group, the age-weighted estimate likely contains some information, but we hypothesise that it is less informative.

6 Evaluation

To evaluate the RVU levy exemption, we aim to assess which parts of the exemption users had jobs associated with each aspect of demanding work and whether they are over- or underrepresented compared to both the employed population and the exemption-eligible population. To this end, we use the predictions and estimated prediction errors of Section 5. We perform a benchmark analysis under the assumption of nondifferentiality and investigate how the results change when this assumption is relaxed.

6.1 Prediction rates

First, we investigate the response predictions for each question, without accounting for prediction errors. As these rates are not yet corrected for misclassification, they correspond to the naive estimate α_y defined in Eq. 3. Figure 7 shows the prediction rates for each group (2) - (4). These rates coincide with the realized fractions of individuals that lie in $\hat{A}_0$ and $\hat{A}_1$, as visualised in Figure 4.

(a) Irregular hours (b) Physically demanding (agree)

(c) Physically demanding (often) (d) Heavy lifting (e) Kneel / stoop

Figure 7: Prediction rates for each question (a) - (e) for the exempted (2), employed (3), and eligible (4) comparison groups. A value of one indicates that the predicted answer is "often" to questions (a), (c), (d), and (e), or "agree" or "strongly agree" to question (b).

Figure 7 shows that for each question, less than half of the individuals are predicted to have demanding work. This is in line with the observed class imbalance in Table 4. For questions (a), (b), and (e), the group of exemption users (2) has relatively more predictions for demanding work than the other groups. For (b), the eligible (4) and exempted (2) groups have similar rates, but higher than those of the employed population (3). However, these figures have not yet corrected for misclassification and correspond to the naive estimate $\hat{a}_y$.

6.2 Benchmark analysis under nondifferentiality

As we hypothesise that the estimates in Table 5 are somewhat informative about the true misclassification rates for the prediction groups, we perform a benchmark analysis under the assumption of transportability. That is, we make the, unlikely, assumption that the unweighted estimates, $\theta_{\hat{x}|x,\text{all}}$, are transportable to the employed group (3) and that age-weighted estimates, $\theta_{\hat{x}|x,\text{age}}$, are transportable to the exempted (2) and eligible (4) groups.

For this benchmark analysis, we compute corrected estimate $\hat{\alpha}_y$ as in Eq. 8, and $\hat{\Psi}_c$ as in Eq. 9, where $c \in \{\text{age}, \text{all}\}$ denotes the control group used for $Y = 0$. Moreover, the variances of these estimates, assuming independence, are estimated using the variance estimators in Eq. 10 and 11, and 95% confidence intervals are computed. The resulting estimates, conditional on the validity of transportability of misclassification rates, are reported in Table 6.

Q	$\hat{\alpha}_{0,\text{all}}$	$\hat{\alpha}_{0,\text{age}}$	$\hat{\alpha}_1$	$\hat{\Psi}_{\text{all}}$	$\hat{\Psi}_{\text{age}}$
(a)	0.19	0.22	0.44	3.32	2.75
	[0.16, 0.22]	[0.15, 0.29]	[0.37, 0.50]	[2.41, 4.58]	[1.94, 3.89]
(b)	0.11	0.10	0.34	3.94	4.77
	[0.08, 0.14]	[-0.01, 0.20]	[0.26, 0.41]	[2.54, 6.11]	[1.74, 13.13]
(c)	0.29	0.24	0.28	0.99	1.24
	[0.26, 0.31]	[0.18, 0.30]	[0.24, 0.33]	[0.77, 1.28]	[0.97, 1.60]
(d)	0.16	0.14	0.24	1.65	1.91
	[0.13, 0.19]	[0.08, 0.21]	[0.19, 0.29]	[1.15, 2.36]	[1.25, 2.92]
(e)	0.10	0.06	0.17	1.87	3.08
	[0.07, 0.12]	[-0.02, 0.14]	[0.11, 0.22]	[1.15, 3.06]	[1.01, 9.34]

Table 6: Estimates of α and Ψ for each question (a) - (e). $\hat{\alpha}_{1,c}$ shows our estimates of $P(X = 1 \mid Y = 1)$, the probability that an exemption user would have answered "often" to questions (a), (c), (d), and (e), or "agree" or "strongly agree" to question (b), under the assumption of either transportability of misclassification rates. Values between brackets denote 95% confidence intervals.

The benchmark results in Table 6 show that, if nondifferentiality holds, in 2020, of exemption users around

(a) 44% worked irregular hours,

(b) 35% had physically demanding jobs,

(c) 28% had often physically demanding jobs,

(d) 24% needed to lift heavy objects,

(e) 17% needed to kneel or stoop.

Additionally, the estimated odds ratios $\hat{\Psi}$ suggest that individuals who worked irregular hours (a), had physically demanding jobs (b), needed to lift heavy objects (d) and had to kneel or stoop (e) are statistically overrepresented compared to both the full employed population and the exemption eligible population. We do not find evidence that individuals whose job was often physically demanding (c) are overrepresented compared to either control group. This last result is interesting, as the similarity between questions (b) and (c) would suggest similar prevalences, but apparently, the perceived frequency of demand and the presence of demand are represented differently. Once again, it should be stressed that these results strongly depend on the assumption of nondifferentiality, of which we argue the validity is unlikely.

Table 6 also shows some interesting results. The estimates for $\hat{\alpha}_{0,\text{age}}$ for questions (b) and (e) show that the confidence intervals span negative values. Since α denotes a probability, which is restricted to the $[0, 1]$ interval, we argue that this is in line with the warning that the delta method requires care near boundary cases, as mentioned in Section 3.

6.3 Robustness to nondifferentiality assumption

To investigate the robustness of our results against the nondifferentiality assumption for exemption users, we investigate what values the odds ratio would take for different specifications of $\theta_{0|01}$ and $\theta_{1|11}$, respectively the specificity and sensitivity of prediction for exemption users. For this analysis, we still assume transportability of misclassification rates to control samples. Results are shown in Figure 8.

Figure 8 shows the counterfactual estimated odds ratios for different specifications of θ. The blue dot denotes the estimated specificity and sensitivity on the age-weighted test set. If the true, unknown specificity and sensitivity lie above the red line, individuals with $X = 1$ are overrepresented among exemption users. If the true values are below the red line, they are underrepresented. An estimate of $\hat{\Psi} < 0$ indicates that $\hat{\alpha}_1 < 0$, which corresponds to a negative probability. Therefore, we argue that these combinations are unlikely.

We argue that the age-weighted estimate is, to some extent, informative about where the true values could lie, but the degree of informativeness is uncertain. Moreover, we hypothesise that the further the estimates are from the decision boundary of $\Psi = 1$, the more likely it is that $X = 1$ is overrepresented among exemption users. Additionally, the ratio of the areas for which $\Psi > 1$ and $0 < \Psi < 1$ also gives some indication of whether overrepresentation is likely, where a relatively large area for the former implies that relatively many likely combinations would produce $\hat{\Psi} > 1$.

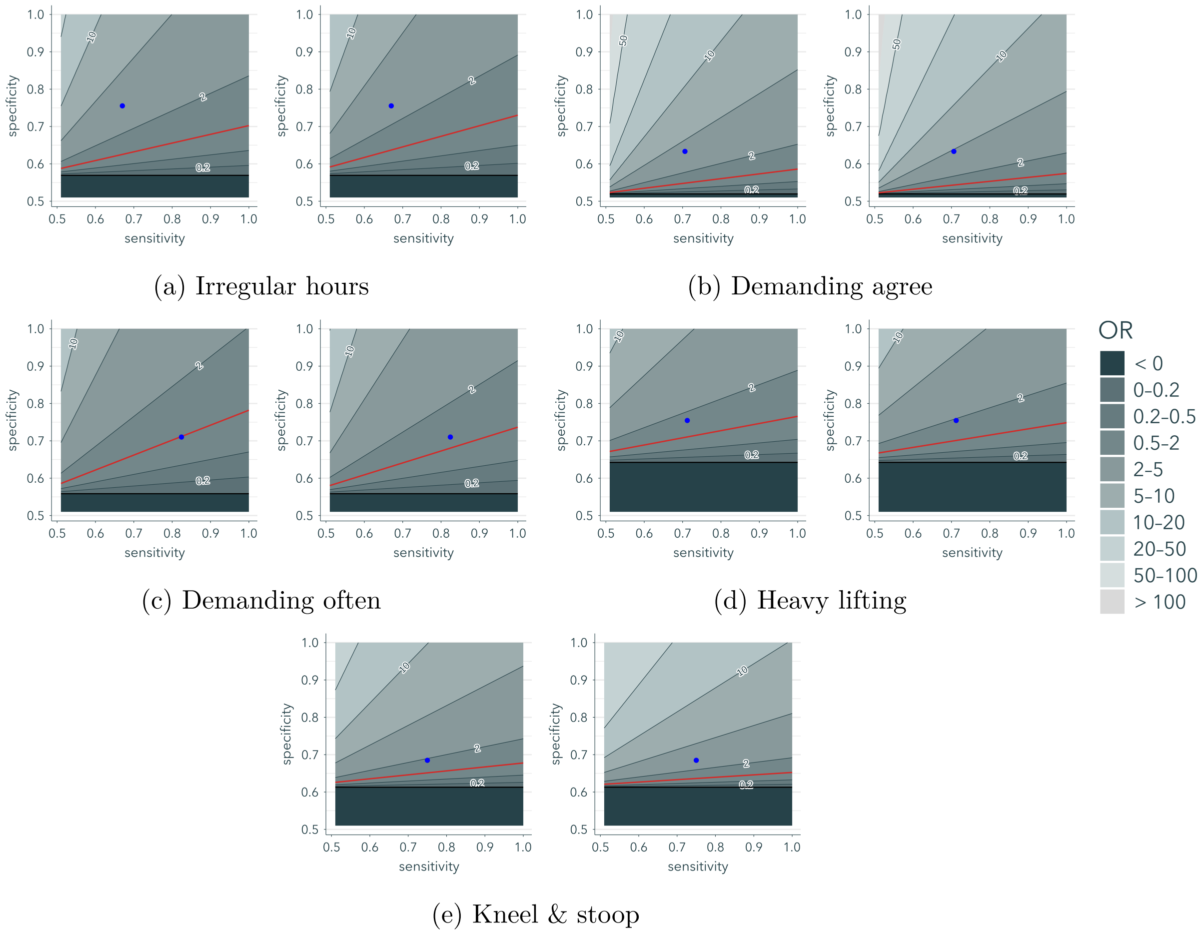


Figure 8: Odds ratios Ψ for different specifications of the specificity and sensitivity of the misclassified demanding work predictions. For each pair of figures, the left evaluates the odds ratio against the employed group (3), and the right evaluates the odds ratio against the exemption-eligible group (4). The red and black lines denote the decision boundary of $\Psi = 1$, and the boundary of $\Psi = 0$, respectively. The blue dot denotes the age-weighted estimates of θ.

It is apparent that, for some questions, the areas associated with under- and overrepresentation cover nearly the full likely set of values. For example, for questions (a) and (b), the true sensitivity and specificity should lie in a small area, relatively far from the age-weighted estimates, for them to be underrepresented. Hence, these results indicate that $X = 1$ is likely overrepresented for these questions. For questions (d) and (e), this effect is less strong. For question (c) the estimates are very close to the $\Psi = 1$ boundary, indicating there is no evidence for either over- or underrepresentation. These results are largely in line with the findings in Table 6.

However, this method is limited in two ways: 1) it still assumes transportability of misclassification rates to the control groups and 2) it does not take into account that some combinations of the sensitivity and specificity are more likely than others, represented by the likelihood of observing a_y given $\alpha_y, \theta_{1|1y}$, and $\theta_{0|0y}$.

6.4 Relaxing the nondifferentiality assumption

To address the two limitations of the robustness analysis, we estimate the posterior probability that $\Psi > 1$, as in Eq. 18. To this end, we evaluate several combinations of prior distributions, where some are informative priors based on the estimates in Table 6. As the transportability of these estimates need not be valid, priors with different degrees of informativeness are considered. This method accounts for both transportability uncertainty and the fact that some parameter combinations are more likely to produce our observations than others, while still using the information provided by test-set estimates.

For the prior distributions $P(\theta_0)$ and $P(\theta_1)$, we specify both an uninformative prior and an informative prior based on the estimates in Table 6, where we estimate $\hat{\Psi}$ for different levels of informativeness, represented by the standard deviation of the prior σ. Because we argue that the control groups more closely resemble a random sample of the population than exemption users, we are more confident that the parameter estimates for controls are closer to the true parameters than those for exemption users. We take this into account for our prior specifications.

When controlling the informativeness of the priors, specifying a smaller standard deviation σ indicates that the prior is more informative. For reference, specifying a Gaussian prior with standard deviation σ indicates that, prior to seeing the data, we are 50% confident that the true values are within $\pm 0.6745\sigma$ and 95% confident that true values are within $\pm 1.96\sigma$.

For $p(\theta_0)$, the prior distribution for control groups $Y = 0$, we make use of three prior-specifications: 1) an uninformative prior uniform over Ψ_p, 2) a strongly informative prior represented by truncated Gaussian distribution with mean and standard deviation as estimated in Table 6, and 3) a weakly informative truncated Gaussian prior with the same mean but the standard deviation 10 times the estimated standard deviation.

For $p(\theta_1)$, we make use of one uninformative and three informative priors with different degrees of informativeness. For the uninformed prior, we specify a uniform distribution over Θ_p. For the informative priors, we specify truncated Gaussian distributions, where the means are equal to the estimates in Table 6, and the standard deviations equal 0.05, 0.10, and 0.20 for the strongly-, medium-, and weakly informed priors, respectively. Here, the most and least informative priors indicate that we are approximately 95% confident that the true values lie within ± 0.10 and ± 0.40 of the age-weighted estimates, respectively, spanning a large portion of the combination in the possible space $1 < \theta_{1|1} + \theta_{0|0} < 2$. Resulting posterior probabilities for $\Psi > 1$ based on $10,000,000$ draws can be found in Table 7.

Table 7 shows that the posterior probability significantly depends on the informativeness of the specified priors, the control group, and the question. The first column, with both uninformative priors for θ_0 and θ_1, shows that the posterior probabilities are close to 0.5, with some slightly lower and most slightly higher. As these values depend only on the observed a_y, they closely relate to the prediction rates for $X = 1$, where higher values indicate that demanding work is

C	Q	$p(\theta_0)$	uninf.	weakly informed				strongly informed			
		$p(\theta_1)$	uninf.	uninf.	weak	medium	strong	uninf.	weak	medium	strong
all	(a)		0.564	0.813	0.910	0.977	0.993	0.822	0.932	0.997	1.000
	(b)		0.516	0.896	0.899	0.957	0.986	0.909	0.931	0.993	1.000
	(c)		0.501	0.714	0.540	0.487	0.477	0.725	0.563	0.506	0.484
	(d)		0.476	0.808	0.719	0.741	0.787	0.825	0.761	0.830	0.963
	(e)		0.480	0.887	0.766	0.723	0.759	0.906	0.819	0.821	0.939
age	(a)		0.564	0.687	0.720	0.752	0.776	0.797	0.905	0.991	1.000
	(b)		0.531	0.728	0.628	0.636	0.643	0.928	0.943	0.994	1.000
	(c)		0.501	0.694	0.509	0.485	0.484	0.777	0.669	0.704	0.825
	(d)		0.599	0.694	0.565	0.532	0.541	0.852	0.798	0.871	0.979
	(e)		0.535	0.767	0.530	0.441	0.434	0.946	0.891	0.908	0.983

Table 7: Bayesian posterior probabilities for $\Psi > 1$ for questions (a) - (e), under uninformed, weakly- and strongly informed specifications of prior $p(\theta_0)$ for the employed and exemption eligible control groups, and uninformed, and weakly, medium and strongly informed priors $p(\theta_1)$ for the exempted group.

overrepresented among exemption users.

For the weakly informed control prior $p(\theta_0)$, the results strongly depend on the informativeness of exemption prior $p(\theta_1)$. For uninformed $p(\theta_1)$, the probability of overrepresentation is generally between $0.7 - 0.9$, indicating that for all aspects, demanding work is likely overrepresented among exemption users, but not statistically significant. As the informativeness of $p(\theta_1)$ increases, posterior probabilities increase for questions (a) and (b), but generally decrease for (c), (d), and (e). This effect is observed in both control groups. For medium and strongly informed priors, posterior probabilities for responses to questions (a) and (b) exceed 0.95.

For the strongly informed prior θ_0, all values are closer to 1. With uninformative $p(\theta_1)$, they are all higher than 0.7. Interestingly, most values increase with a more informed prior $p(\theta_1)$, but only question (c) with the employed control group decreases. Question (c) against eligible, and questions (d) and (e) against both decrease compared to uninformed $p(\theta_0)$, but increase as the prior gets more informative. Additionally, we see many significant values, especially for the strongly informed $p(\theta_1)$ and for questions (a) and (b).

Interestingly, the differences between control groups appear larger for the weakly informed $p(\theta_0)$ than the strongly informed alternative. Particularly for question (e), the informativeness of $p(\theta_0)$ makes a big difference. The estimates for the strongly informed $p(\theta_0)$ and strongly informed $p(\theta_1)$ seem to approximately coincide with our observations in Figure 8.

In general, the results in Table 7 seem to coincide with the results in Table 6. However, because the results differ substantially across columns, conclusions depend on the confidence one has in the transportability of prediction errors to both the exemption group and the control groups.

7 Conclusion and discussion

This thesis aimed to assess whether the RVU levy exemption targeted employees with demanding jobs in the period 2021-2024. To this end, we assessed the probability that exemption users had jobs related to several aspects of demanding work, and whether they were over- or underrepresented compared to the employed and exemption-eligible populations. To this end, we predicted responses to several questions in the LISS Work & Schooling survey, using individual-level features in the CBS microdata. In order to perform inference using these predictions, however, assumptions have to be made about the transportability of misclassification rates from the test set to exemption and control group predictions.

A benchmark analysis shows that, if the assumption of nondifferentiality were valid, a relatively large share of exemption users did work associated with aspects of demandingness, and individuals who had demanding work are overrepresented among exemption users, both relative to the employed population and the exemption-eligible population. Especially employees who worked irregular hours and agreed they had physically demanding jobs appear overrepresented. Interestingly, employees who perceived their jobs to be demanding **often** do not appear to be overrepresented.

Using test set estimates as informative priors allows us to relax the assumption of nondifferentiality and compute Bayesian posterior probabilities that our measures of demanding work are overrepresented among RVU users. The resulting probabilities indicate again that employees who worked irregular hours and those who agree they had demanding jobs are likely overrepresented. However, these conclusions strongly depend on how informative the estimated misclassification rates are for predicting the exemption and control groups, which cannot be determined.

Hence, we conclude that the RVU levy exemption appears to target employees whose job is demanding in terms of working hours, whereas for employees whose work is demanding in terms of physical load, the results depend on the specific aspect of the physical load. However, this conclusion requires careful consideration, as using predicted survey responses inherently relies on restrictive assumptions about how well misclassification rates of predictions can be estimated based on available responses. To make confident claims about the performance of the RVU levy exemption, further research is required.

7.1 Discussion

In addition to the general downsides of using response predictions discussed throughout this thesis and shortcomings in its implementation, this study has several limitations.

This thesis is unable to fully evaluate the effectiveness and efficiency of the RVU levy exemption during the period 2021-2024. In this study, we focus on the fraction of exemption users who had demanding work and to whether they are over- or underrepresented, but the case-control setting prevents us from evaluating the fraction of employees with demanding work who took an RVU. Also, this study does not investigate the causal effect of taking an RVU on early retirement, nor does it consider cost-efficiency.

The survey questions considered in this research do not cover all load fields distinguished by TNO. Hence, our conclusions only regard the "working times" and "physical load" fields proposed by TNO, and do not provide a general overview on how other aspects of demanding work are represented among RVU users.

Regarding the data used, our measures of demanding work, the LISS survey responses, are self-reported and therefore likely to contain measurement error. This is not taken into account explicitly in our research. Additionally, the dataset of exemption users does not include the exact year the RVU was taken, which prevents us from making year-before-retirement estimates. Moreover, the data appear noisy, as they likely do not provide a perfect measure of RVU levy exemption usage. This argument is supported by the observation that some individuals who took an RVU did not appear in the employment registration.

7.2 Further research

There remains considerable room for further research on the RVU levy exemption. This thesis aimed to evaluate the exemption for the period 2021-2024, but several regulatory changes have been made since. From 2026 onward, regulators require RVUs to be more targeted and controlled. To this end, clearer guidelines were developed to determine which functions are eligible for an RVU. As data on RVU use in this period are not yet available, evaluating the policy in its current form is left for future research.

Regarding the limited load fields considered in this thesis, questions from other sources, such as the EBB, could be considered that account for other aspects of demanding work, or new measures could be constructed by combining questions from the LISS panel Work & Schooling survey.

A primary problem this research addresses is the lack of a population-wide measure of the extent to which employees have demanding work. If this data were to become available, evaluating the RVU levy exemption would become trivial. Information from a larger subsample could already yield further insights, for example, by providing more confidence in claims about misclassification rates. If misclassification rates could be estimated for a random sample of exemption users, the assumption of nondifferentiality would hold, and reliable conclusions could be drawn.

This thesis lays a strong foundation in methods for using survey response predictions, but there remain many opportunities for further development and improvement. For example, the prediction models produced values between 0 and 1, which were converted to binary indicators using a threshold. We hypothesise that in this process, some information was lost, and that using the raw predictions could be a direction for further exploration, perhaps to improve estimates of sensitivity and specificity. There is also much room for improvement in the prediction model.

A more thorough preliminary analysis of what features correspond to exemption users could already improve prediction performance, but using more informative features could likely yield larger improvements.

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